Cells are the basic structural and functional units of all living organisms. The cell theory states that all living things are composed of cells, that the cell is the smallest unit of life, and that new cells arise only from existing cells. Prokaryotic cells lack a membrane-bound nucleus, while eukaryotic cells keep their DNA inside a nucleus. Bacteria are prokaryotes, whereas plants, animals and fungi are eukaryotes.

The plasma membrane surrounds every cell and controls which substances enter and leave. It is built from a phospholipid bilayer in which proteins are embedded. Because the membrane lets some molecules cross but not others, it is described as selectively permeable.

Mitochondria are the organelles where cellular respiration produces most of the cell's ATP. Cellular respiration breaks down glucose in the presence of oxygen and releases carbon dioxide and water. Glycolysis is the first stage of respiration and takes place in the cytoplasm. The Krebs cycle occurs in the mitochondrial matrix, and the electron transport chain is located on the inner mitochondrial membrane.

One molecule of glucose yields about 30 to 32 molecules of ATP through aerobic respiration. Glycolysis alone produces a net gain of 2 ATP per glucose molecule. When oxygen is scarce, muscle cells switch to fermentation, which regenerates NAD+ and produces lactate.

Chloroplasts are the organelles that carry out photosynthesis in plant and algal cells. Photosynthesis converts light energy into chemical energy stored in glucose. Chlorophyll is the green pigment that absorbs light, mainly in the blue and red wavelengths. The light-dependent reactions occur in the thylakoid membranes and split water, releasing oxygen. The Calvin cycle takes place in the stroma and fixes carbon dioxide into sugar.

Ribosomes are the sites of protein synthesis. They read messenger RNA and link amino acids together in the order specified by the genetic code. Ribosomes can float freely in the cytoplasm or be attached to the rough endoplasmic reticulum, which gives it its studded appearance.

The cell cycle is the sequence of growth and division that a cell goes through. Interphase is the longest phase, during which the cell grows and copies its DNA. Mitosis divides the nucleus into two genetically identical nuclei and has four stages: prophase, metaphase, anaphase and telophase. Cytokinesis then splits the cytoplasm to form two daughter cells.

Meiosis is a special type of division that produces gametes with half the number of chromosomes. A human body cell contains 46 chromosomes, so a human gamete contains 23 chromosomes. Crossing over during meiosis exchanges genetic material between homologous chromosomes and increases variation.